

Payment Notice

Payment Notice- Request

Payers should send the following details in the request body while making a call for Payment Notice Request to providers:

1. A set of header attributes that provide transport, security, message integrity and summary information about the message being exchanged. This information is used by the NHCX gateway for routing the request and auditing purposes.
2. Domain payload containing the PaymentNotice Request as specified for the use case by the domain specifications in the link.
<https://nrces.in/preview/ndhm/fhir/r4/StructureDefinition-PaymentNotice.html>
3. This needs to be encrypted so that NHCX cannot read this and can be decrypted & processed only by the intended recipient.
4. Request body (header attributes and the FHIR bundle) should be sent in the form of a JWE token ([RFC-7516](#)) using the steps defined in NHCX specs.

The response to this API could be one of the following:

1. A successful accepted response from the NHCX gateway if the structure of the request is valid and the validation of open attributes (protocol headers) is successful. Upon successful validation, NHCX gateway forwards the same request to the intended recipient asynchronously.
2. An error response if any of the validations fail.

Payment Notification– Response

This is the callback API on NHCX gateway and on Payer systems which will be called by Provider systems for payment acknowledgement and NHCX gateways to return the response for Payment Notification requests.

Payload for this API has to be created as per the specifications of NHCX published in the link <https://nrces.in/preview/ndhm/fhir/r4/StructureDefinitionPaymentNotice.html>

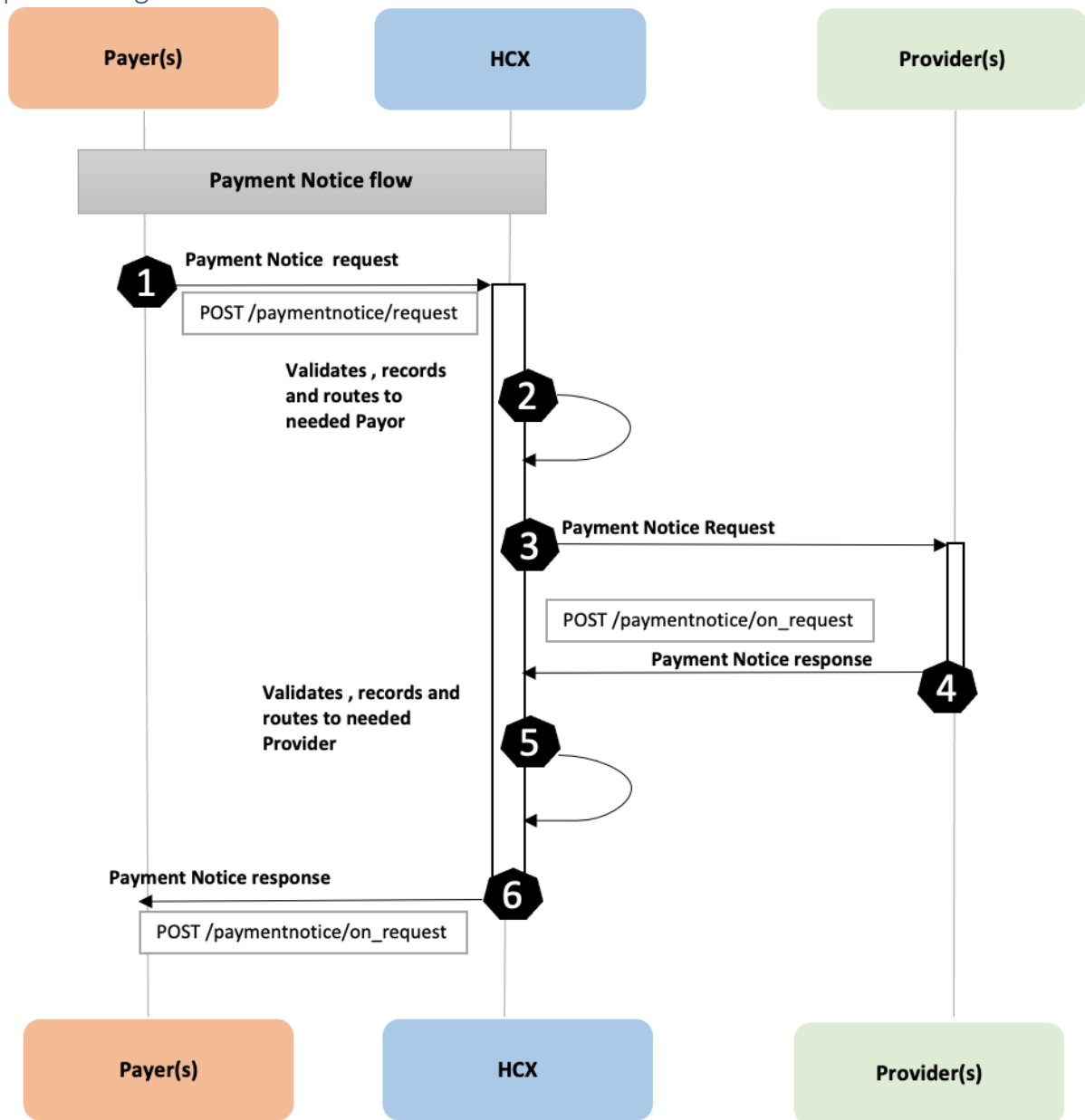
1. A set of header attributes that provide transport, security, message integrity and summary information about the message being exchanged. This information is used by the NHCX gateway for routing the request and auditing purposes.
2. If payment notification request is successfully processed, provider should send a Payment Notification domain entity as specified for the use case in domain specifications in the link.
<https://nrces.in/preview/ndhm/fhir/r4/StructureDefinition-PaymentNotice.html>

3. This needs to be encrypted so that NHCX cannot read this, and this can be decrypted & processed only by the intended recipient.
4. In any scenario if Payer is unable to process the request due to Provider is invalid or decryption of the message fails or any mandatory protocol attributes are missing with regards to the use case, an error message will be sent as “protocol response” which contains the same protocol header attributes along with error messages. The error details should be sent in the following manner: - Basic details about the error must be sent as part of the header attributes. NHCX gateway shall read and store this information for audit & search purposes.
5. Following header attributes should be used for sending the error details:
 - a ****x-hcx-status**** should be set to ****response.fail****.
 - b ****x-hcx-error_details**** attribute should be set with the appropriate error code, message, and trace details (if available).
6. In case where there are business related errors, the Payer should send exact details of the error to the recipient. These details should not be shared with the NHCX gateway and hence, should be sent in encrypted form.

Note: Refer to the JWEPayload schema definition for details & structure of the JWE token that must be sent as the request body in this API. The response to this API could be one of the following:

7. A successful accepted response from the NHCX gateway if the structure of the request payload is valid and the validation of open attributes (protocol headers) is successful. Upon successful validation, NHCX gateway forwards the same request to the intended recipient asynchronously.
8. An error response if any of the validations fail.

Sequence Diagram



High-level definition of API

API Definition – Payment Notice Request- Provider

Name of the API	/paymentnotice/request
Description/Purpose	This API is for Payors to send Payment notification/reconciliation objects to Providers via the NHCX gateway. This API is available on NHCX gateways and Provider systems.

	Payload for this API must be created as per the Payment Notice Bundle defined in NHCX Specifications and serialized as JWE json as per the guidelines in NHCX Specifications.
Input (API to be consumed by)	Payers
Key Processing/Validations	Validate the protected header values
HTTP Method	POST
External API	
Input JSON Structure	As per API specifications published.
Response JSON Structure	As per API specifications published
HTTP Response Status Code	202- Accepted
Exception Code	400 – Request Validation Failed 404- Request resource was not found 500 – Downstream systems down/unhandled exception
Error Response JSON	As per API specifications published
Remarks (If Any)	

API Definition – Payment Notice Request - Payer

Name of the API	/paymentnotice/on_request
Description/Purpose	This is callback API for payment acknowledgement by the Providers. This API is available on NHCX gateways and Payor systems. Payload for this API must be created as per the Payment Notice Bundle defined in NHCX Specifications and serialized as JWE json as per the guidelines in NHCX Specifications.
Input (API to be consumed by)	Providers
Key Processing/Validations	Validate the protected header values
HTTP Method	POST
External API	
Input JSON Structure	As per API specifications published
Response JSON Structure	As per API specifications published
HTTP Response Status Code	202- Accepted

Exception Code	400 – Request Validation Failed 404- Request resource was not found 500 – Downstream systems down/unhandled exception
Error Response JSON	As per API specifications published
Remarks (If Any)	